

Mitec 10GHz transverter modifications

- This document is based on the notes of David VK3QM and Chas VK3PY who took copious notes during the modifications of their systems. Without their work many systems would not be converted.

- 1990's era 10GHz microwave links
- Generally turn up with 600mm dish
- Separate TX / RX blocks
- High / low power TX versions
- High / low gain RX versions (?)
- Crystal locked
- 640MHz uplink / 70MHz downlink *

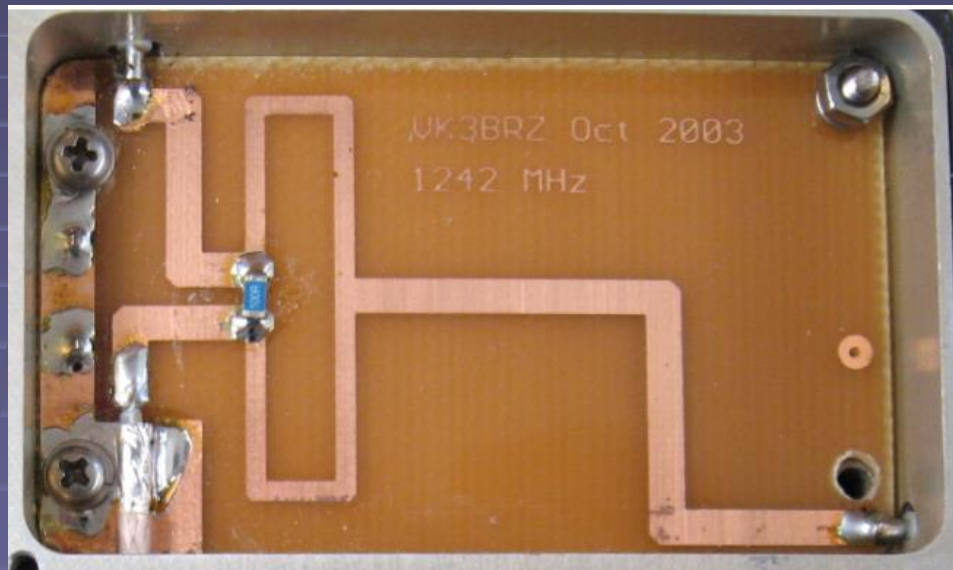
- RX modifications
 - Disable crystal bank
 - Remove 1.2GHz PLL
 - Remove demodulator
 - Downconverter changes
 - 1200MHz LO modifications

- TX modifications
 - Disable crystal bank
 - Remove 1.2GHz PLL
 - Remove modulator
 - Upconverter changes
 - 1200MHz LO modifications

Snowflake tuning

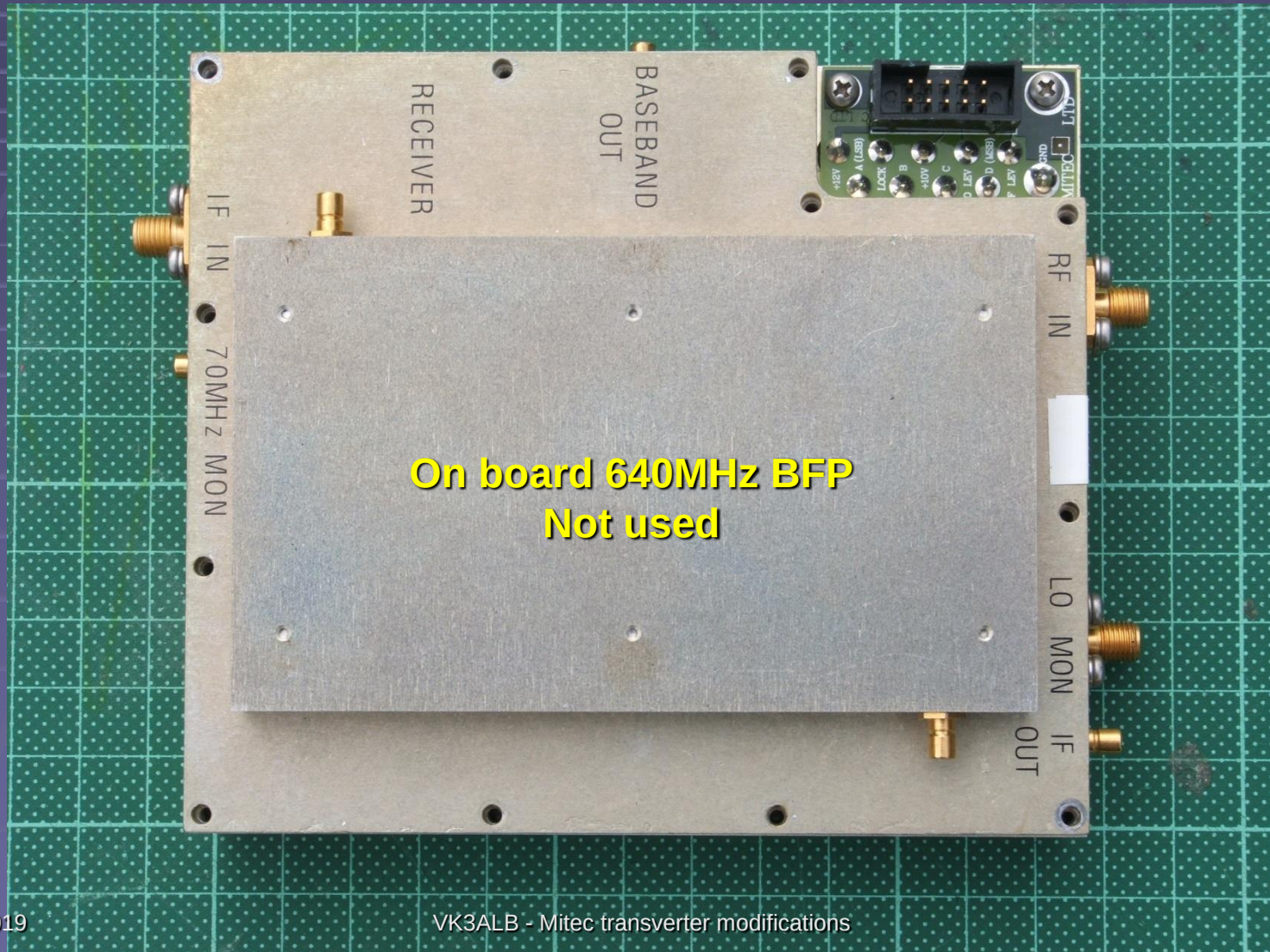
- You will need to use the dark art of snowflake tuning to tune the RX & TX chains for best results. It's easier to do than explain but very important.
- This requires placing small bits of copper along the signal path to increase the level. It's experimental and no two systems will be the same.

- LO modifications
 - VK3UHF group use 1200MHz splitter to feed both TX & RX modules

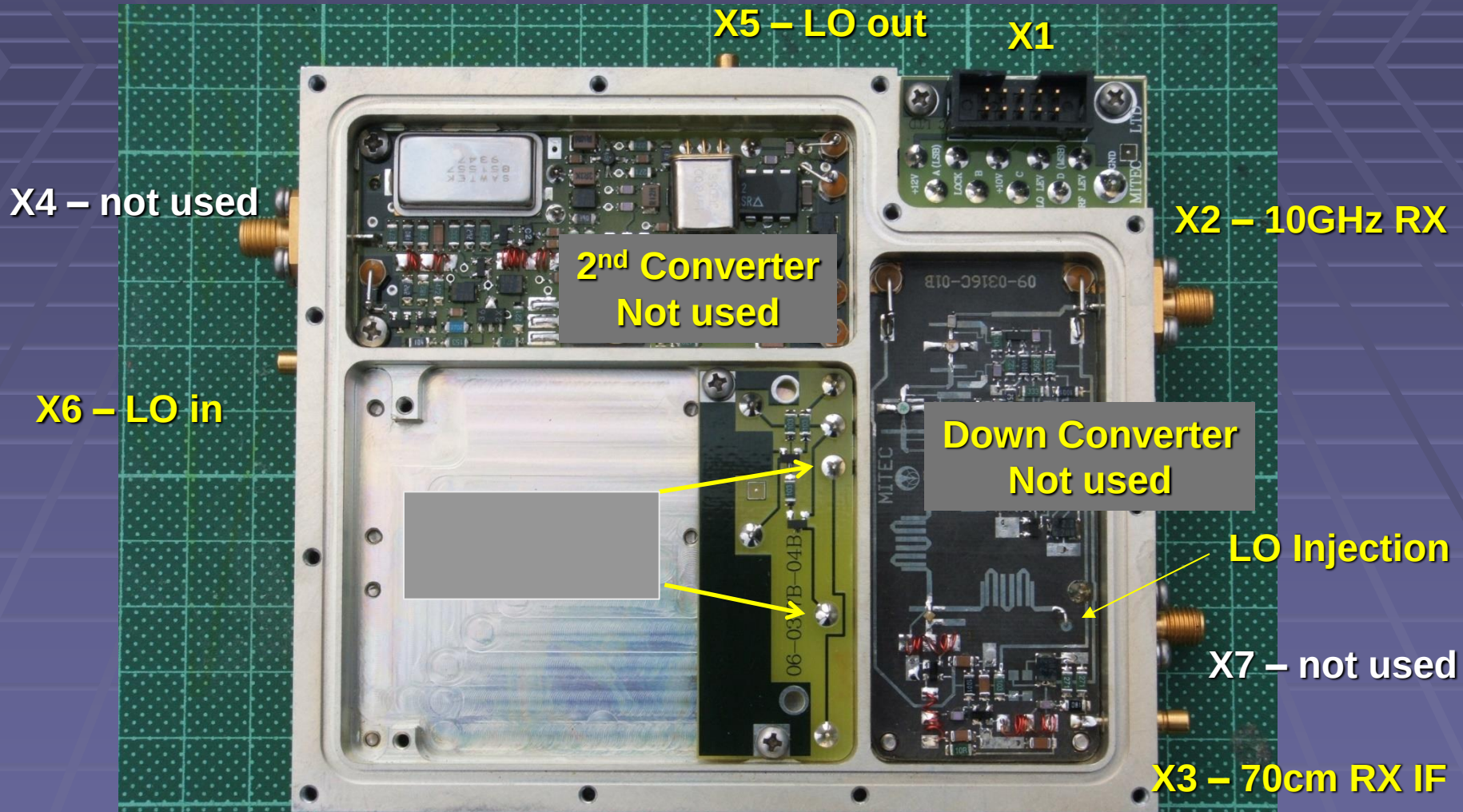


- There are other options for splitting LO.

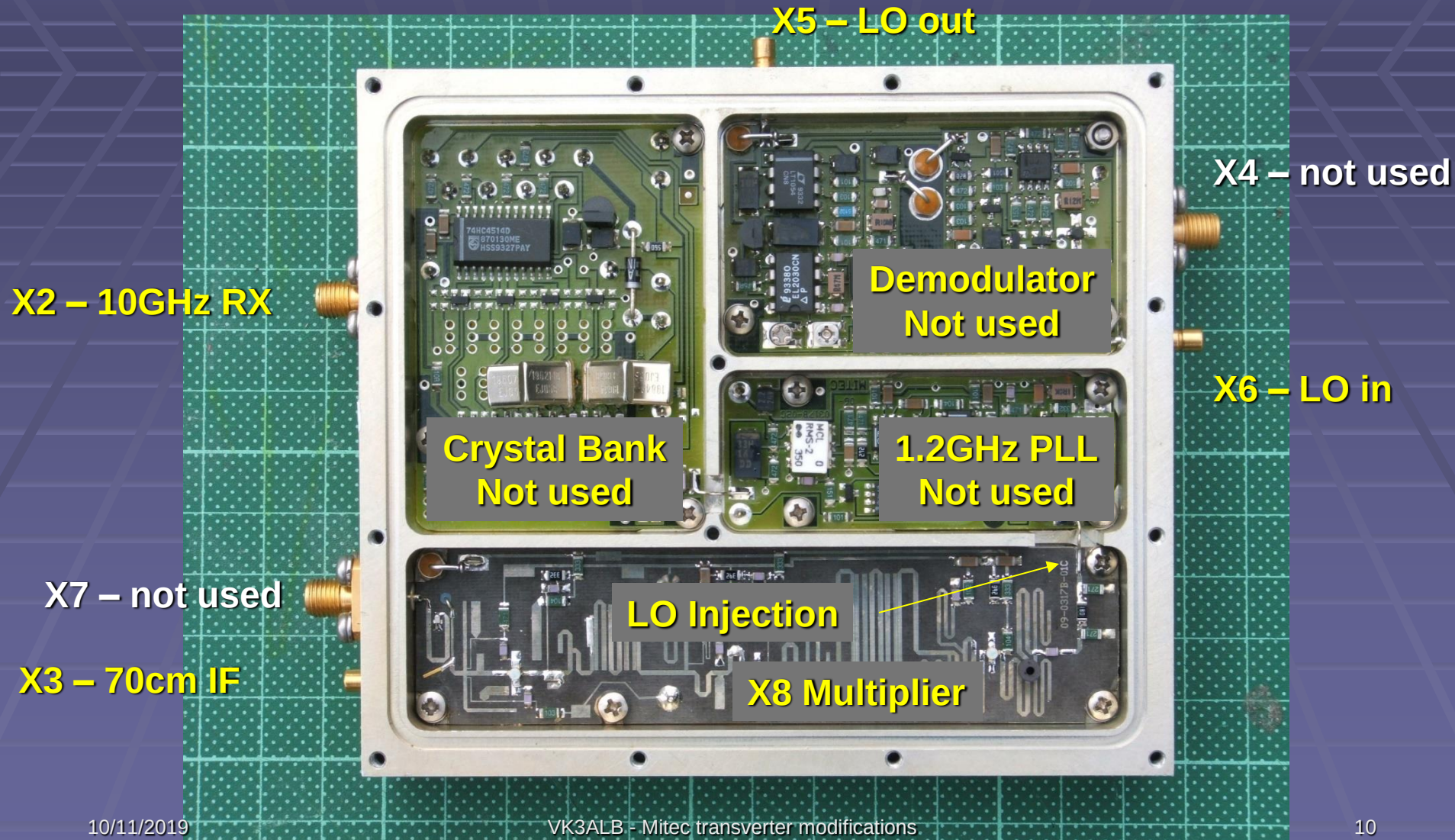
RX module



RX module



RX module



RX & TX Crystal Banks – A1

RF Lev

LO Lev

Lock

+12V Supply

GND on
reverse side
of board.

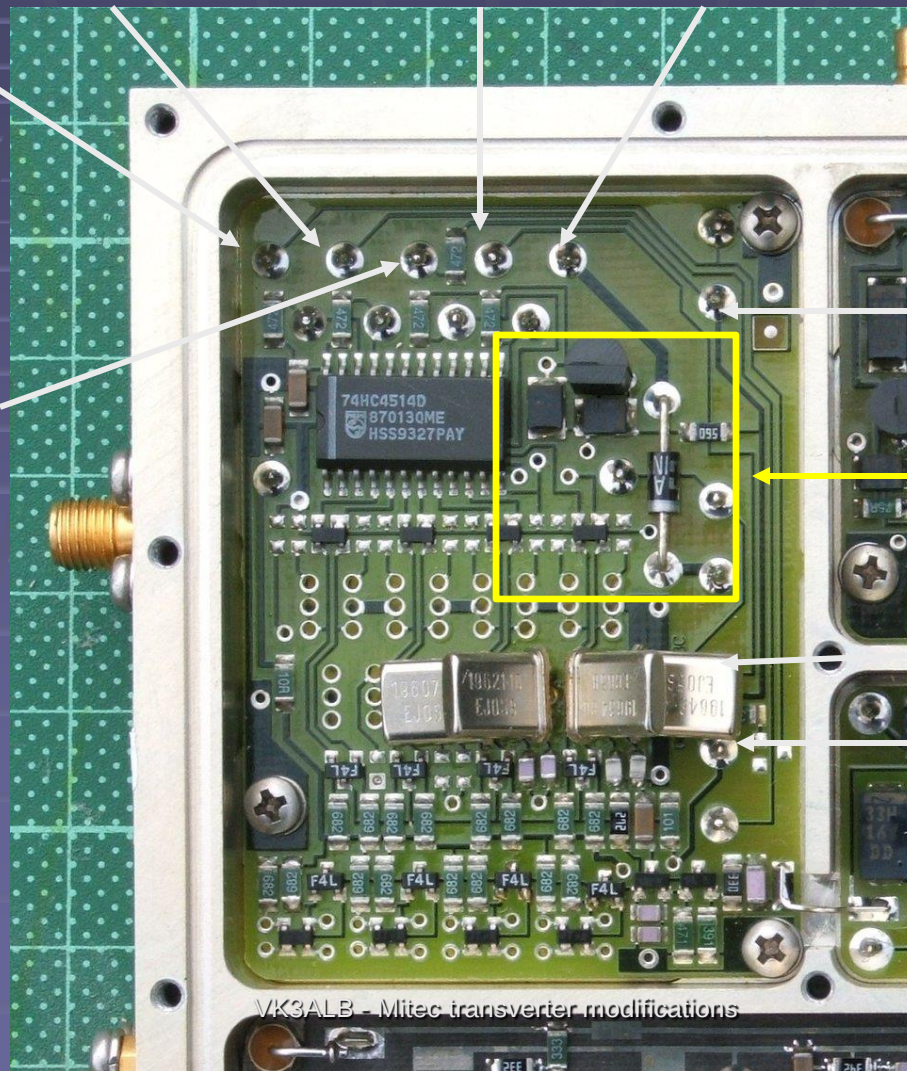
+10V TP

+10V Supply

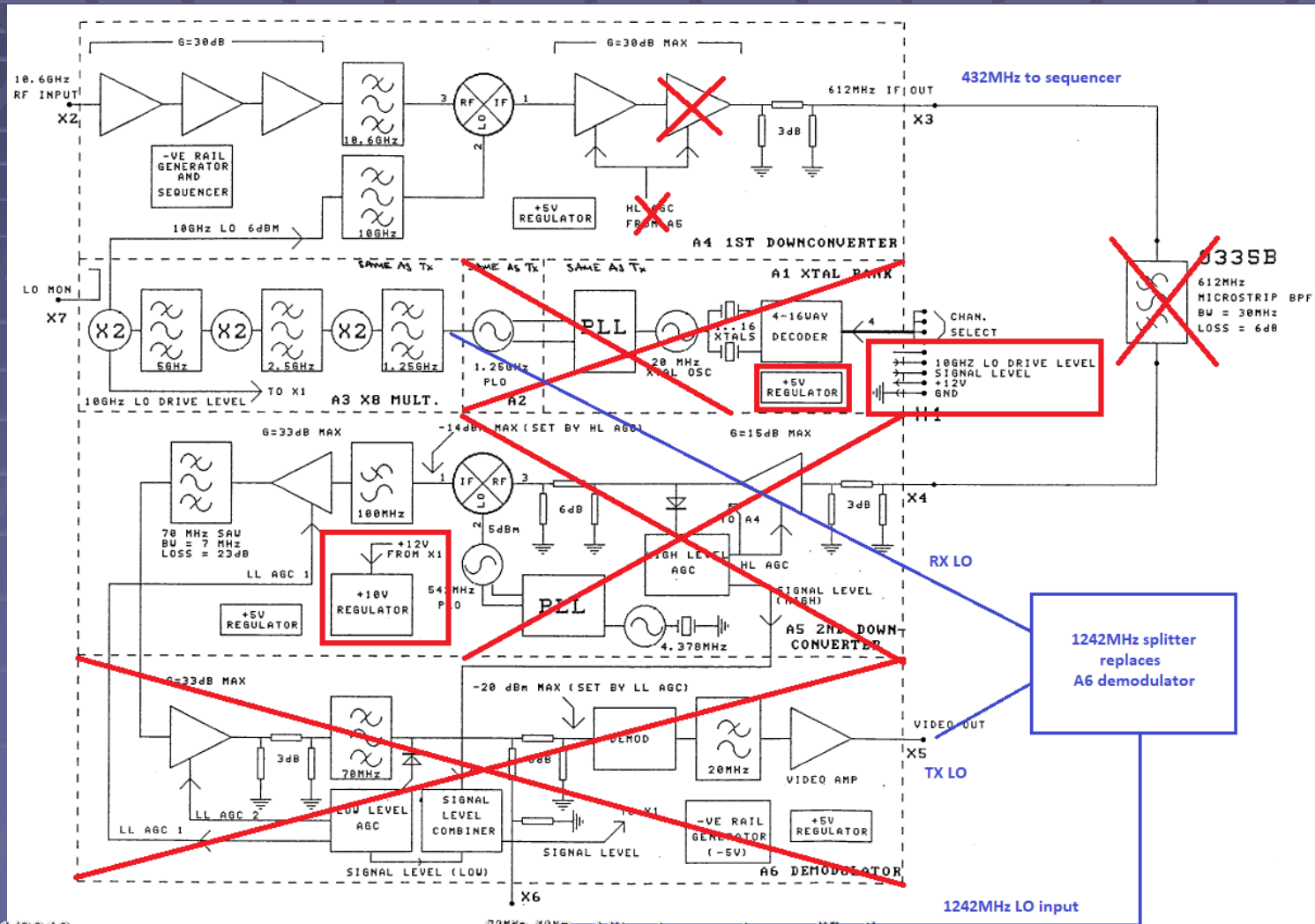
**Retain these
components**

Lock

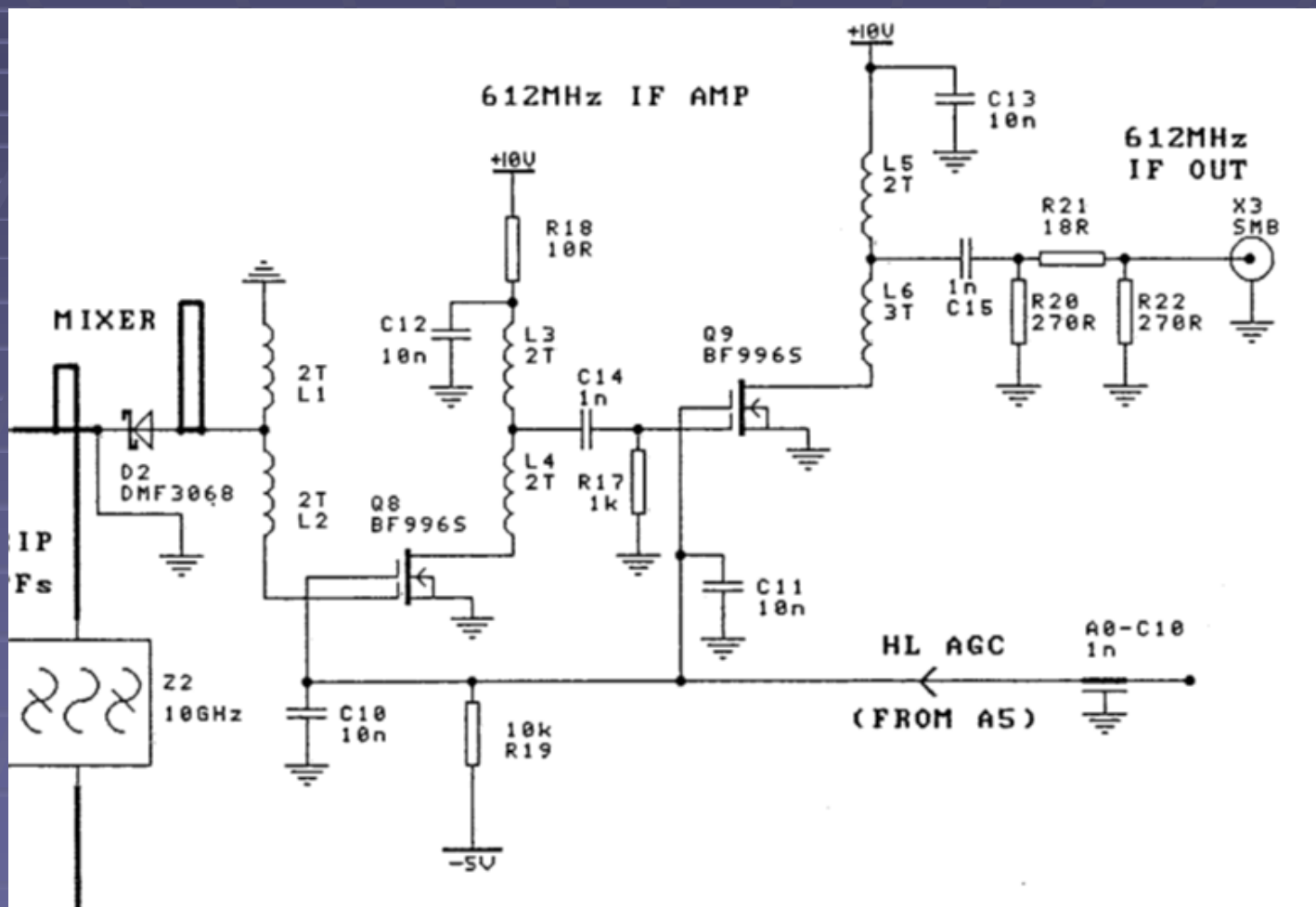
+10V Supply



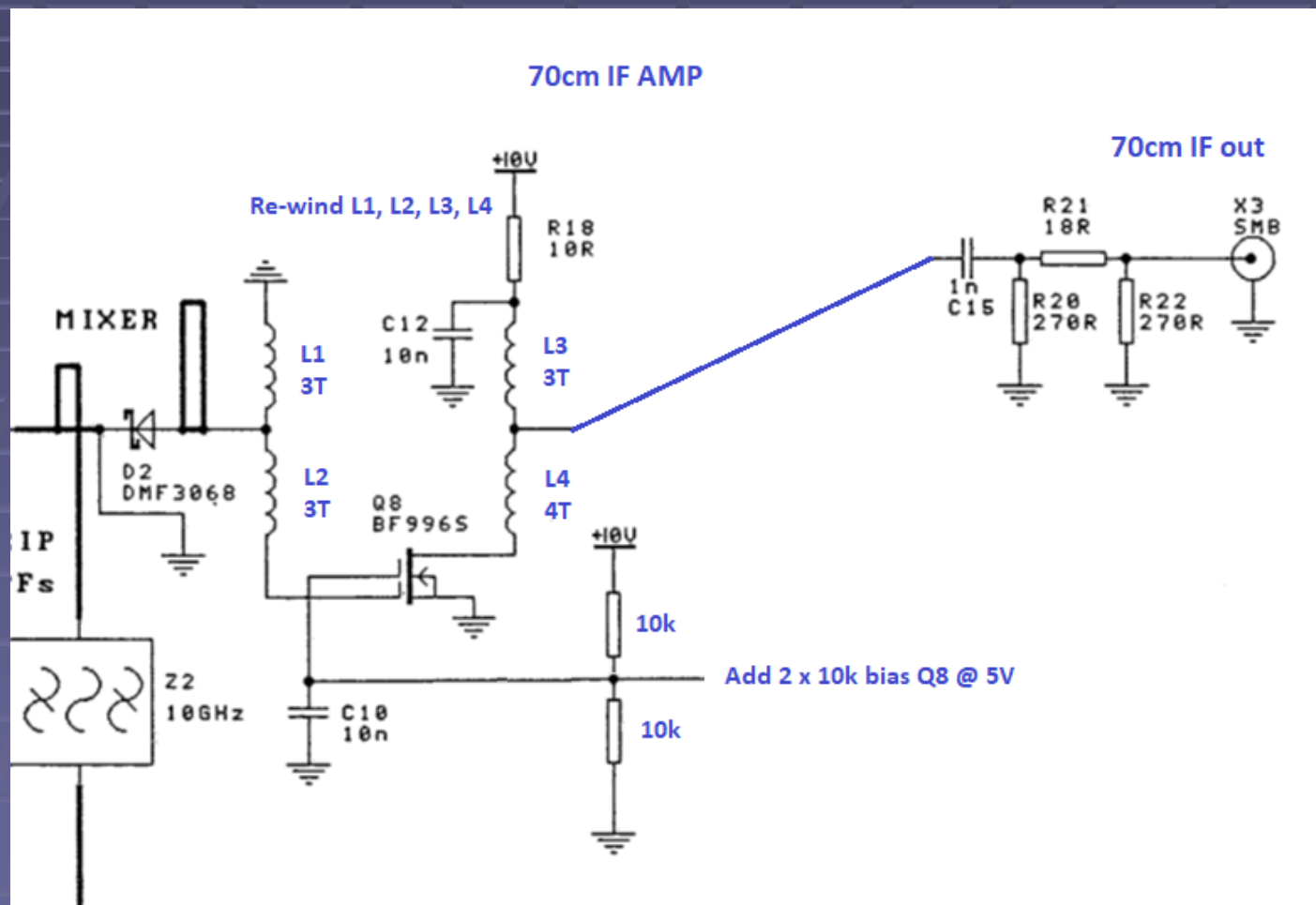
RX module



RX Down Converter - A4



RX Down Converter - A4

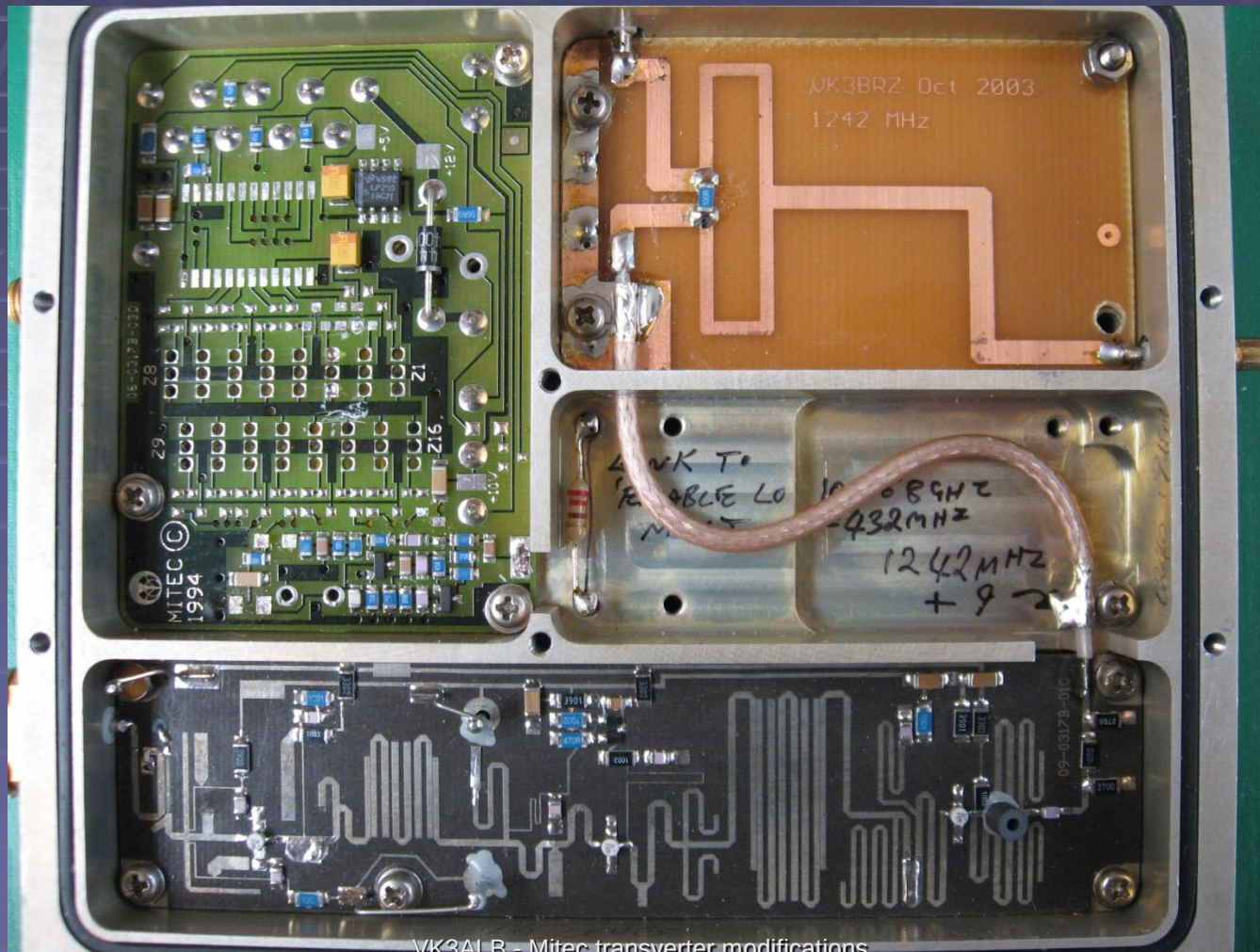


RX Down Converter - A4



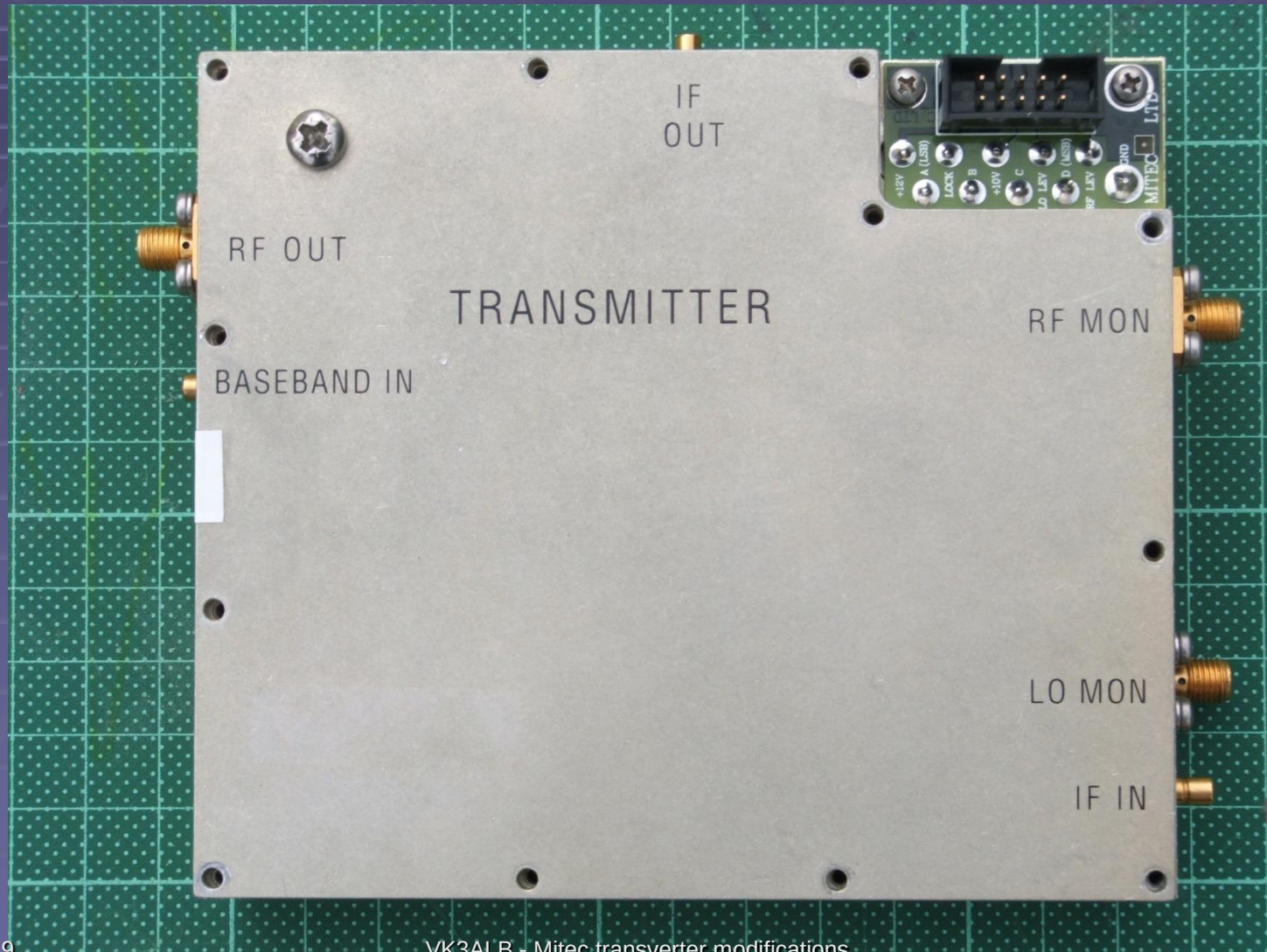
RX LO conversion

X5 – LO out to TX module

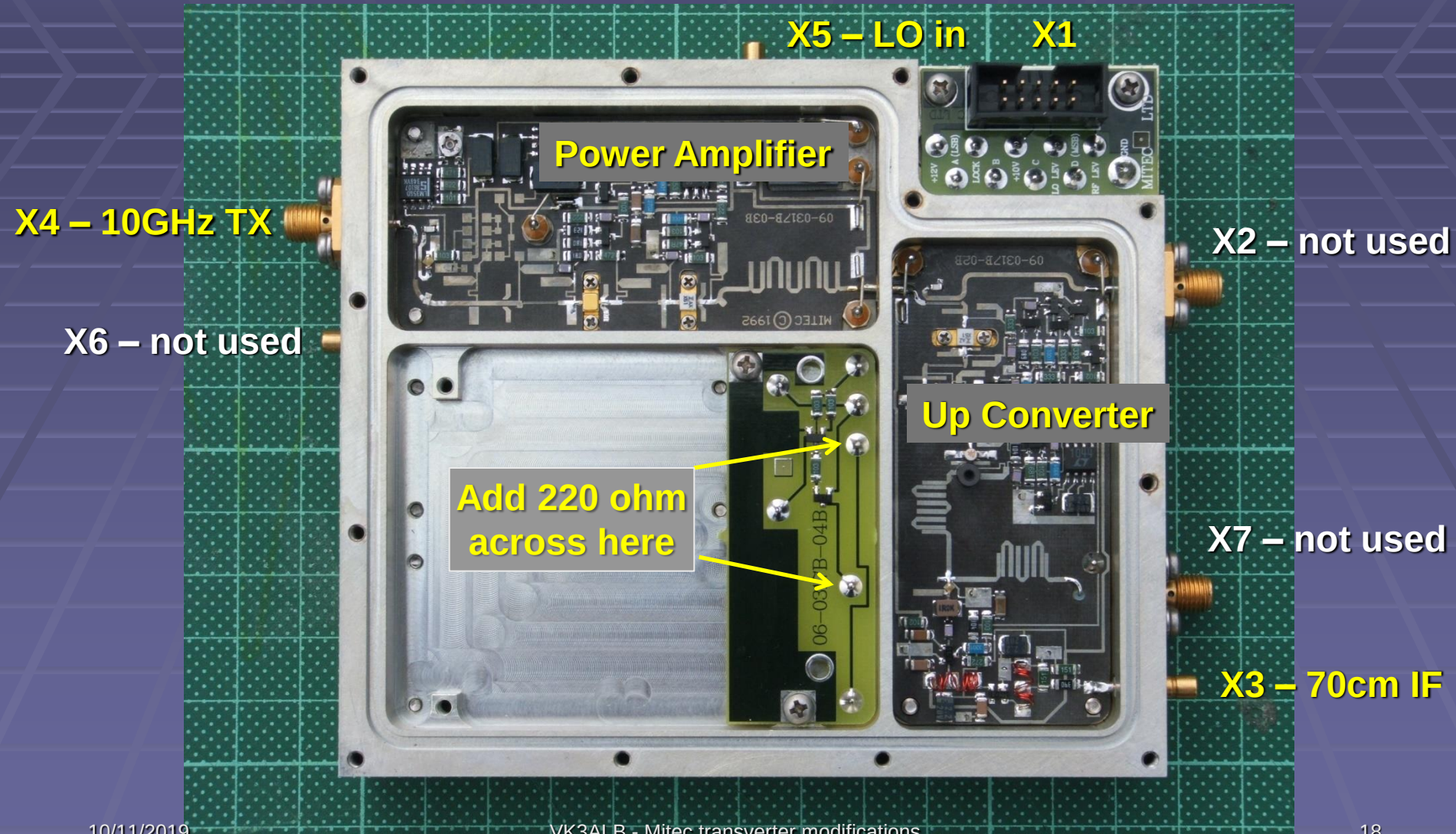


X6 – LO in

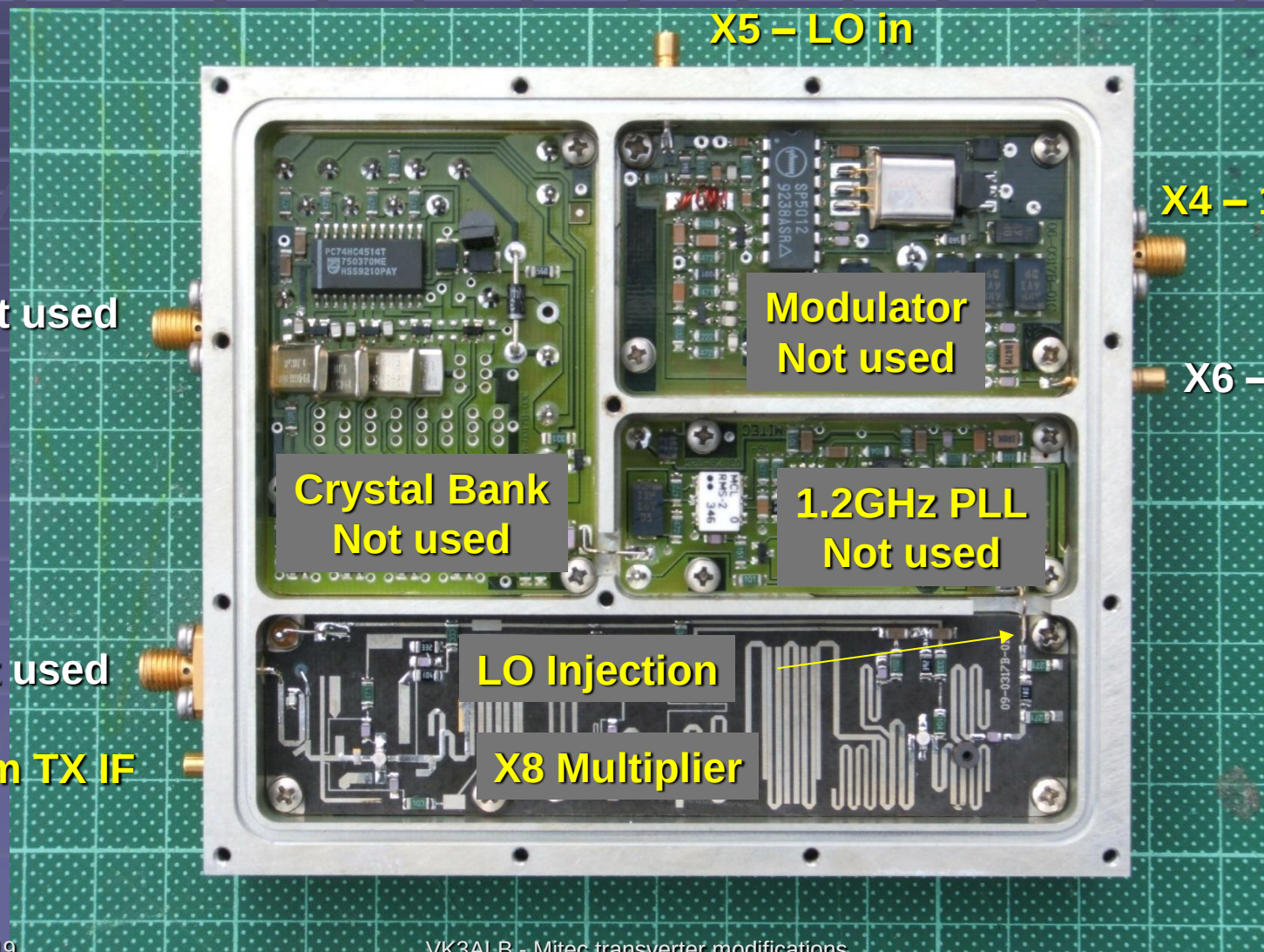
TX module



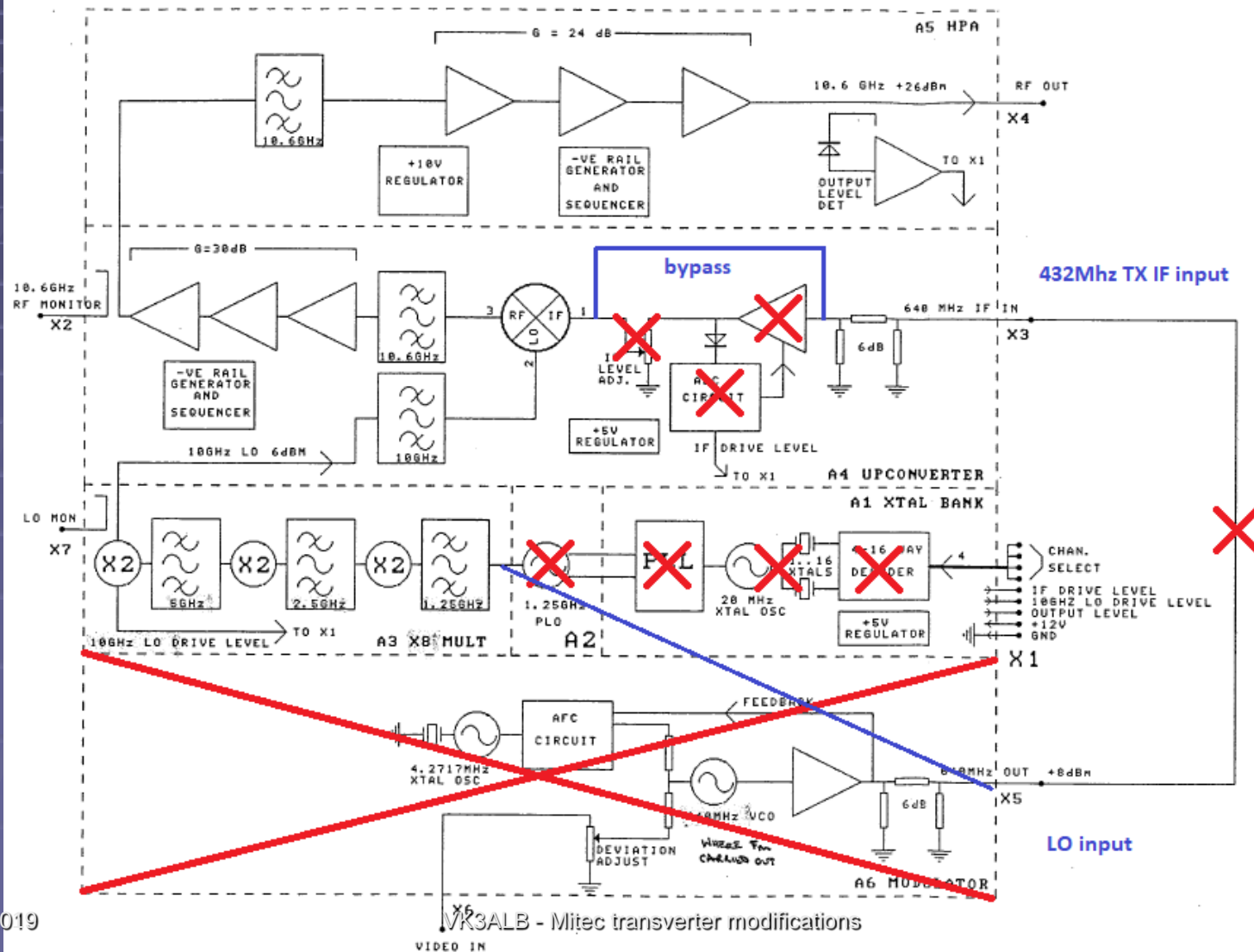
TX module



TX module



TX module

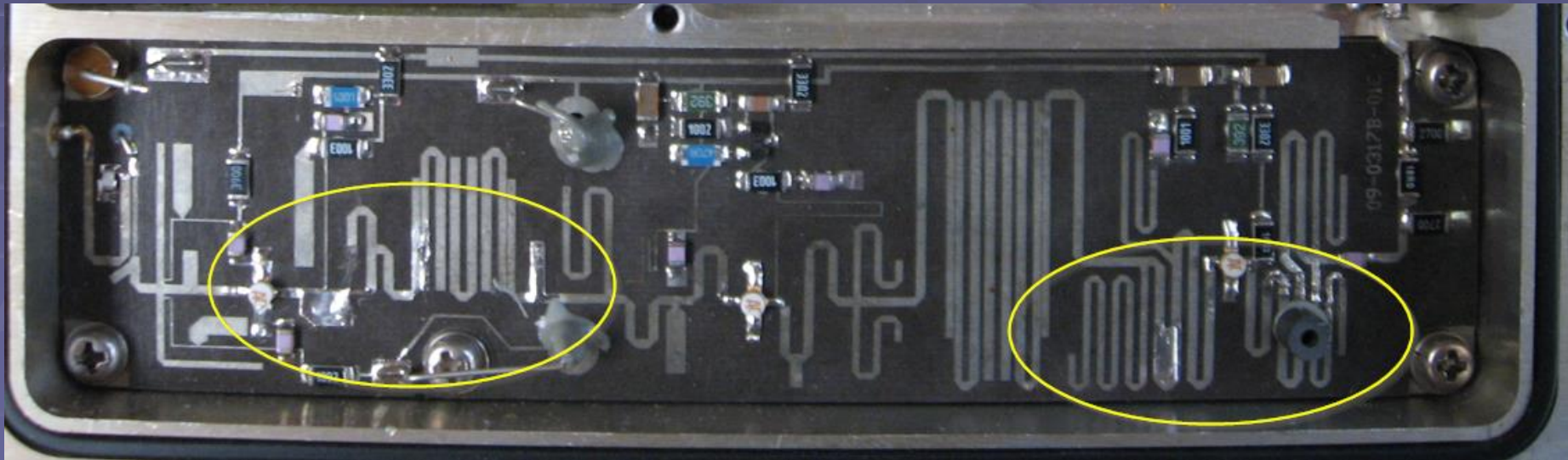


TX LO conversion



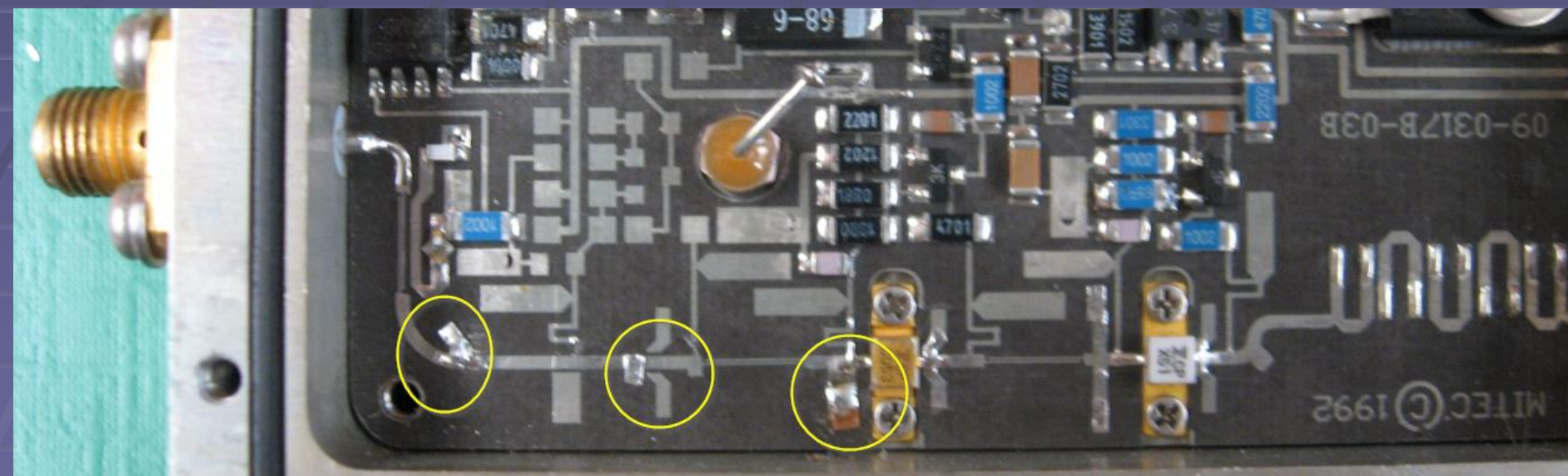
TX tuning

- Example of TX snowflaking
- May not be the same on all systems

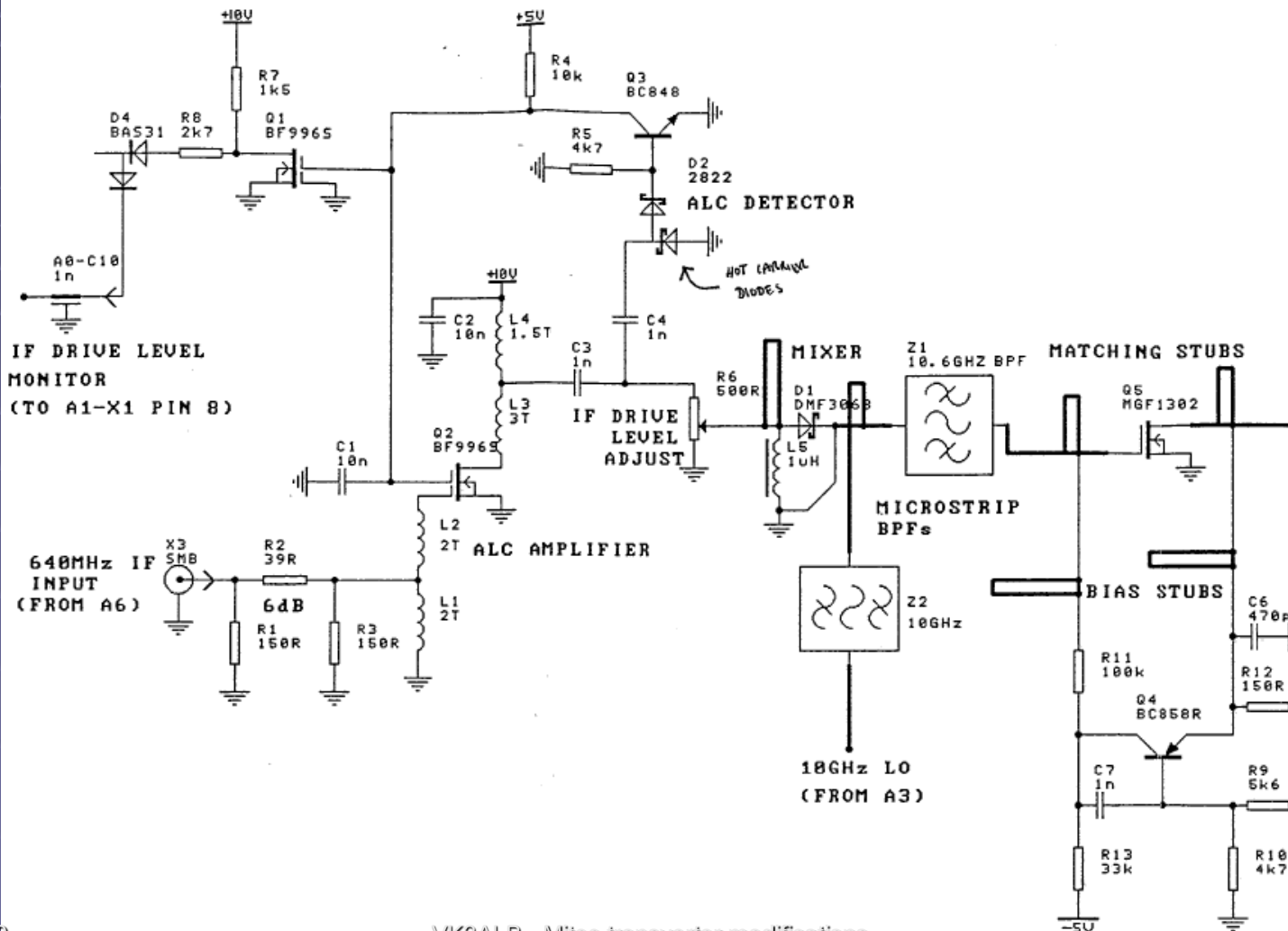


TX tuning

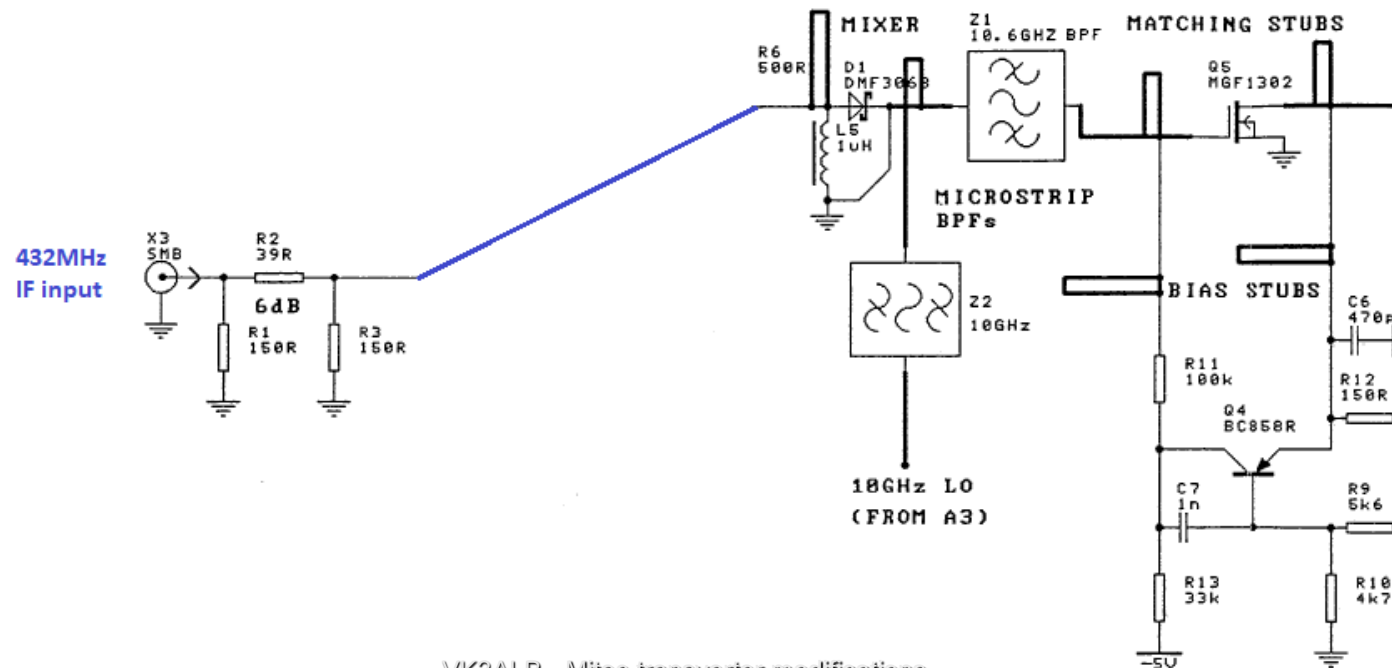
- Example of TX snowflaking
- May not be the same on all systems



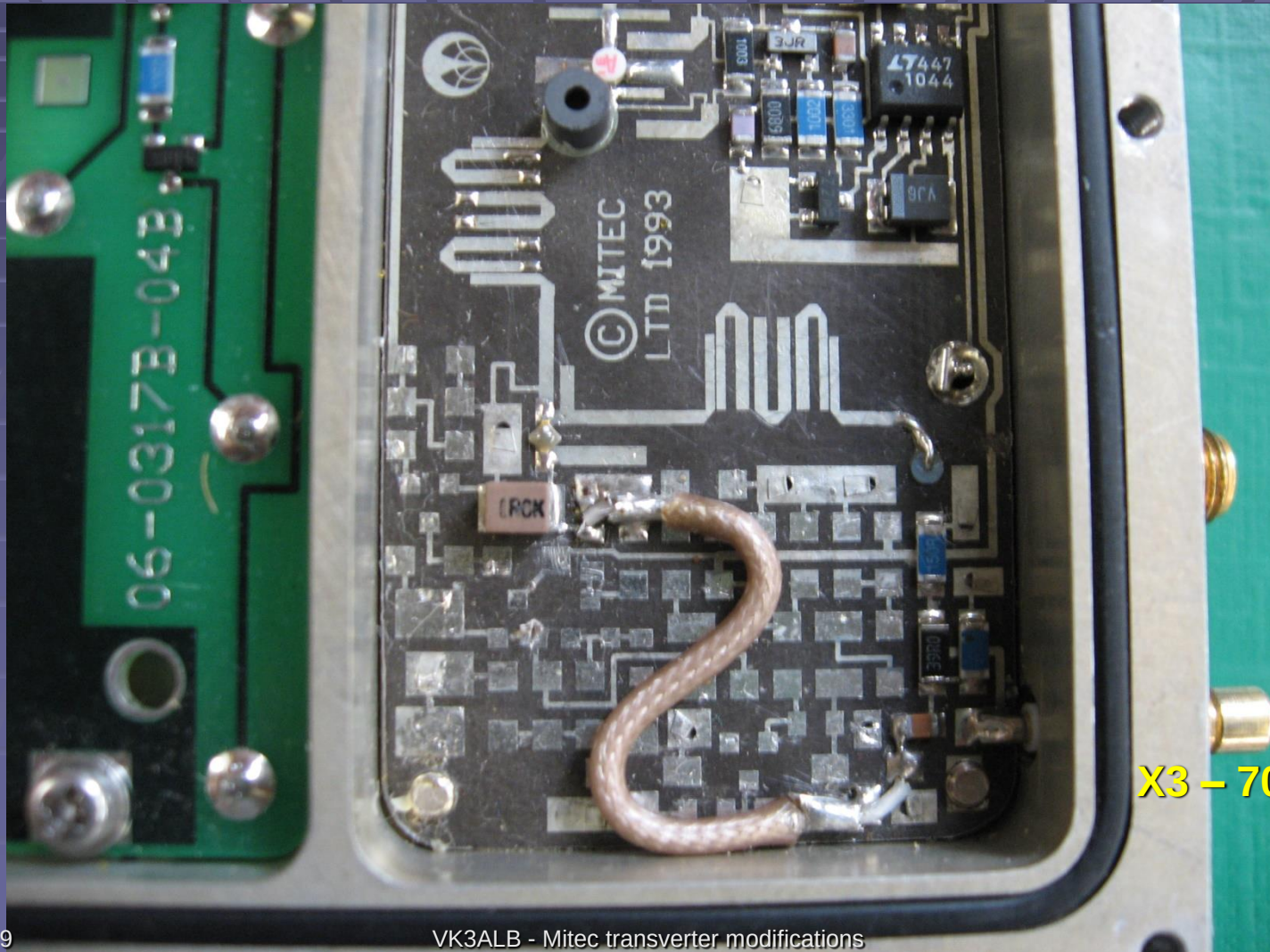
TX up-converter



TX up-converter

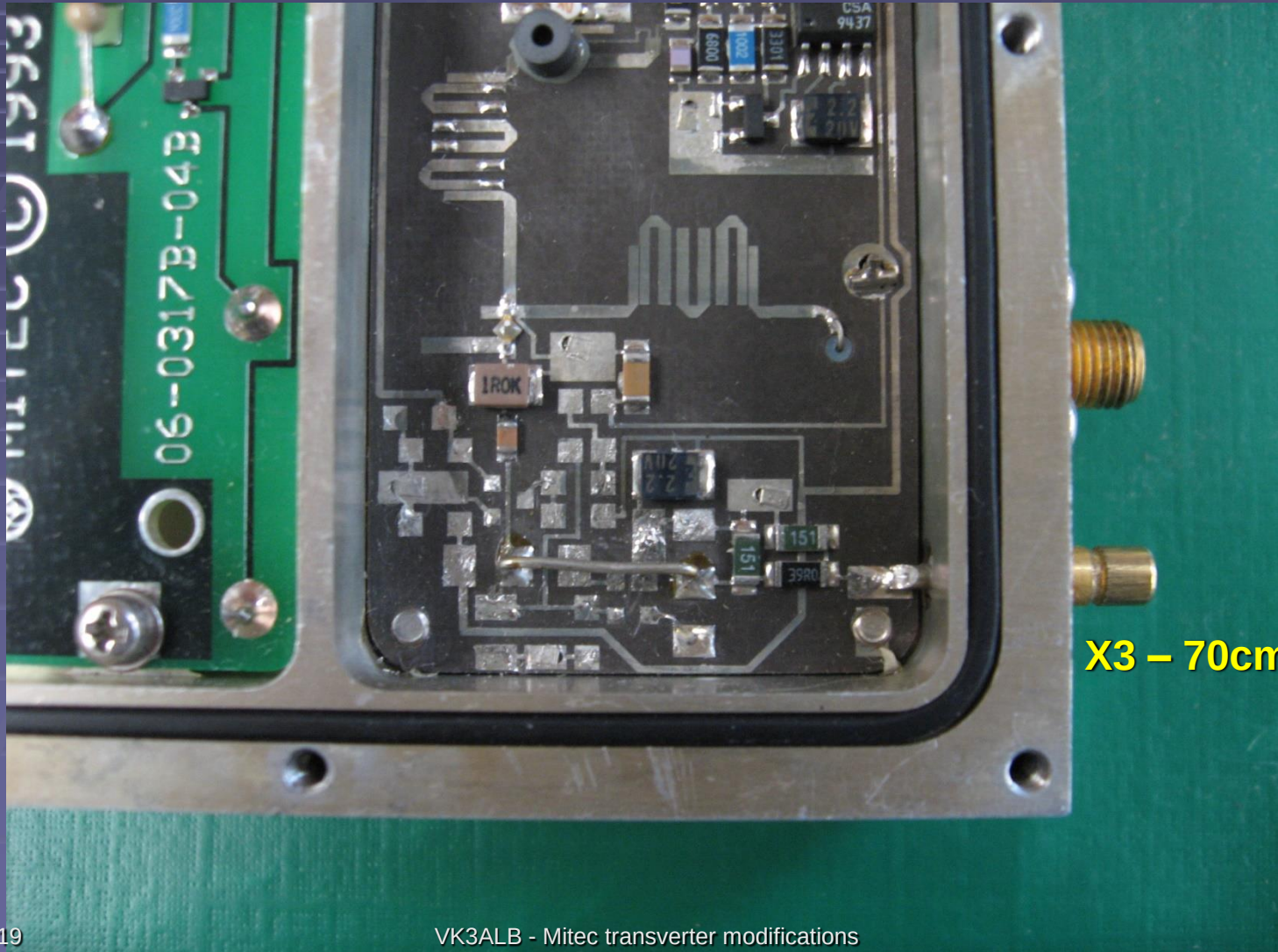


TX IF modification



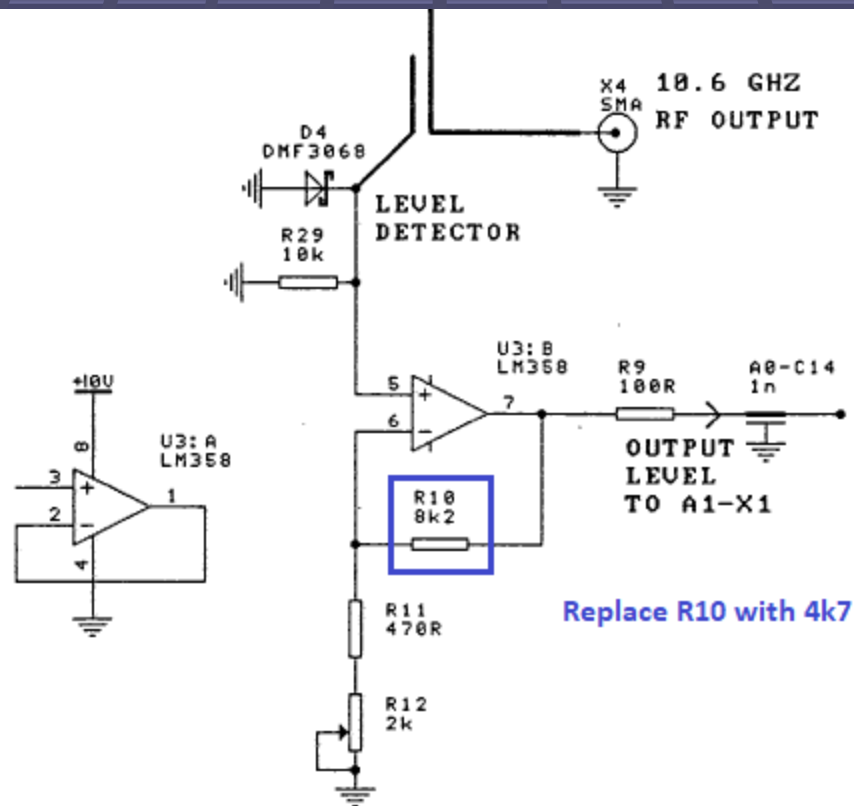
X3 – 70cm TX IF

TX IF modification

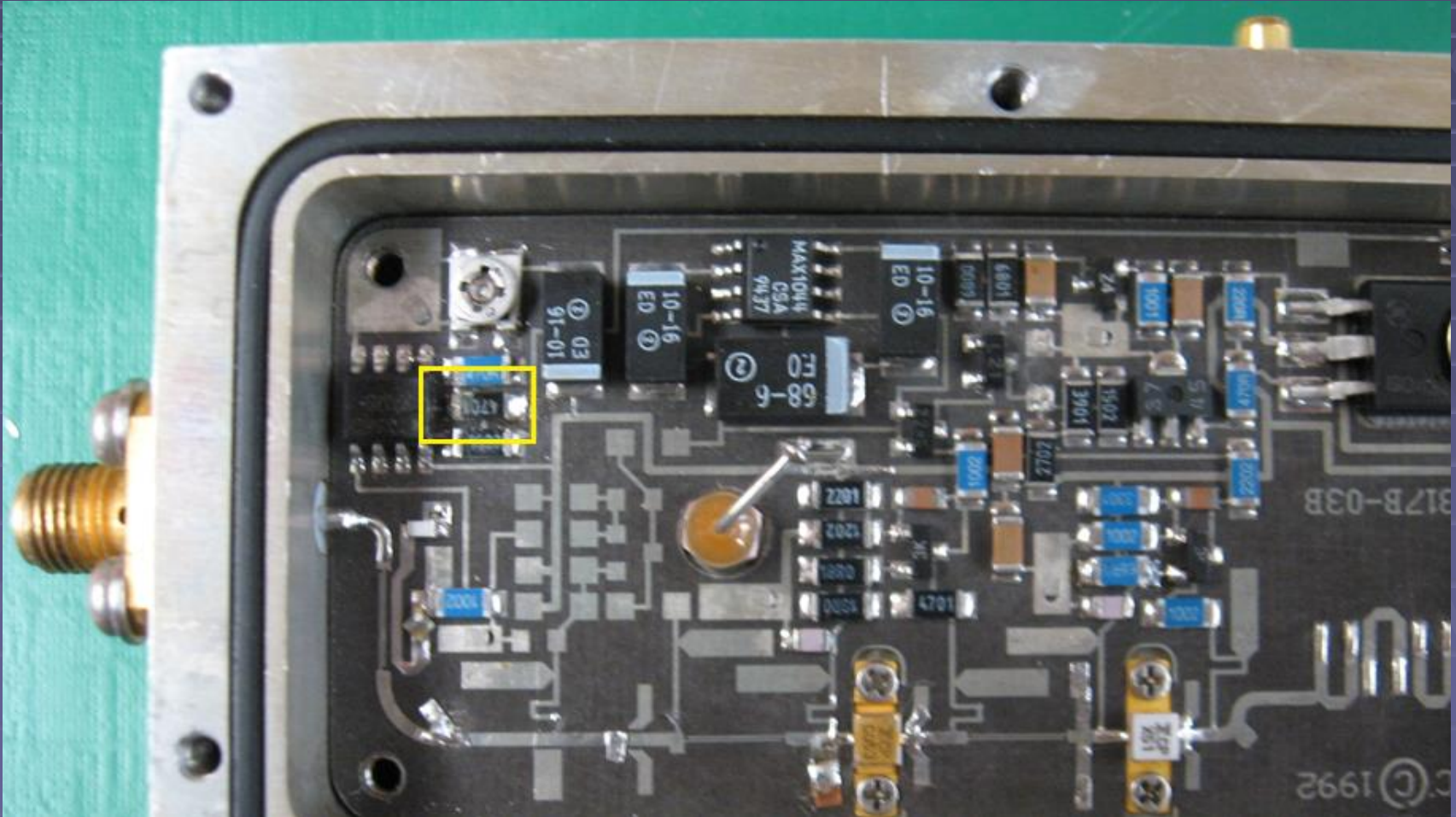


X3 – 70cm TX IF

TX power monitor



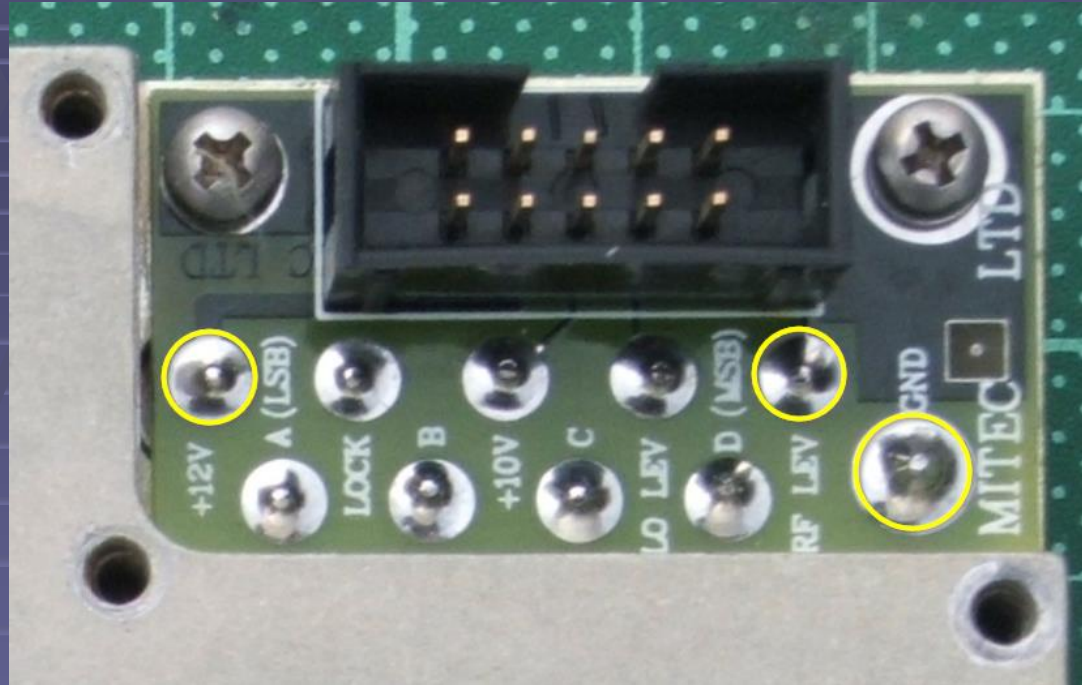
TX power monitor



Putting it all together

- Power supply – nominal 13.8V
- LO requirements, min +10 dBm
 - $10368 - 432 / 8 = 1242\text{MHz}$
 - $10368 - 433 / 8 = 1241.875\text{MHz}$
 - $10368 - 435 / 8 = 1241.625\text{MHz}$
 - Each block requires LO at least +7 dBm
 - Adjust on test for best TX/RX performance
- TX IF injection level – adjust on test
- TX power monitor
 - DC level proportional to TX power

Putting it all together

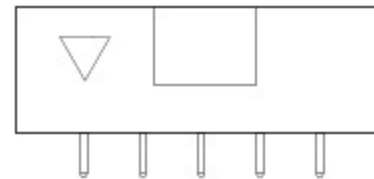
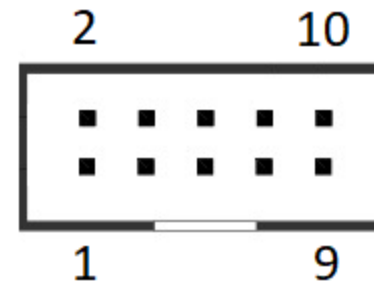


- Solder connections for both modules
- Really only need GND, +12V and RF Lev for TX if you have a meter.

Putting it all together

■ X1 connections

X1	RX module	TX module
1	Ground	Ch sel D *
2	Ch sel A *	Ch sel C *
3	Ch sel B *	Ch sel B *
4	Ch sel C *	Ch sel A *
5	Ch sel D *	Ground
6	+12 volts	TX RF power
7	NC.	TX LO drive *
8	RX LO drive *	Mon IF drive *
9	RX AGC *	+12 volts
10	NC.	NC.

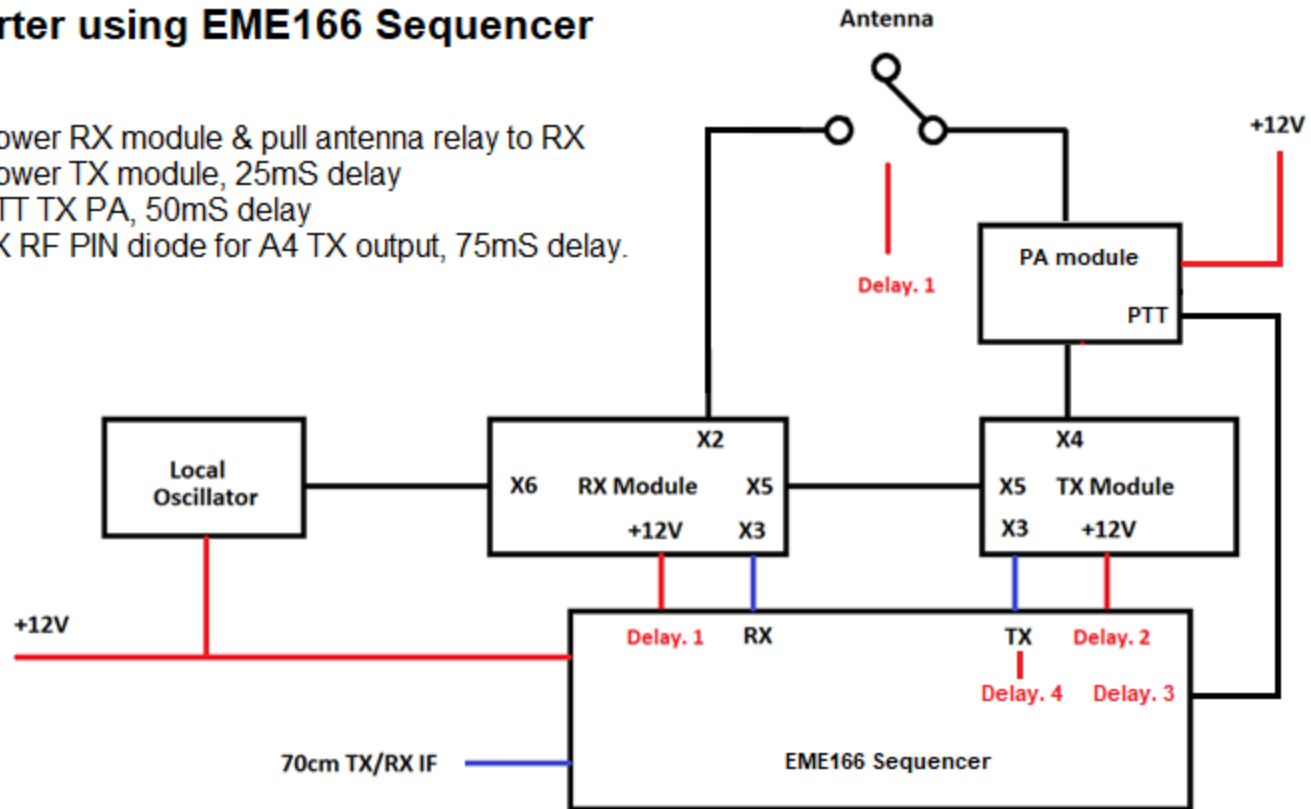


* not used

Putting it all together

Transverter using EME166 Sequencer

- Delay 1 – Power RX module & pull antenna relay to RX
- Delay 2 – Power TX module, 25mS delay
- Delay 3 – PTT TX PA, 50mS delay
- Delay 4 – TX RF PIN diode for A4 TX output, 75mS delay.

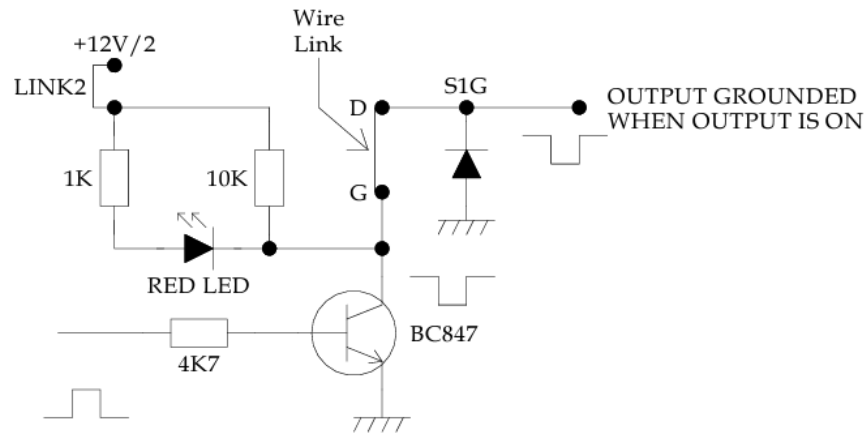


Putting it all together

- Active low PTT line for Amplifier
 - The EME166 board must be modified for active low PTT

EME166 MOD FOR LINEAR AMPLIFIER SWITCHING

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- 1/ Remove the NDT2955 MosFET from the board.
- 2/ Bridge the pads with wire that were originally connected to the MosFET Gate & Drain connections.
- 3/ This mod will allow the switching of Linear Amplifiers that require a grounded input to switch to TX mode.

Documentation and links

- [Website of VK3KBC](#)
- [Notes on snowflaking p5](#)
- [Notes on snowflaking p20](#)
- [Notes on snowflaking p23](#)
- [EME166 PTT modification](#)

Acknowledgements

- VK3UHF – Lara Labs
 - David VK3QM
 - Chas VK3PY
- Ian VK3AXH
 - Proof reading

End